

Decoder-Only Transformers for Cross-Lingual Transfer Learning in Automatic Speech Recognition

Amishi Gangwar



4th Year Project Report
Artificial Intelligence and Computer Science
School of Informatics
University of Edinburgh
2025

Abstract

This dissertation explores the performance and potential of decoder-only transformer architectures for cross-lingual transfer learning in Automatic Speech Recognition (ASR). Although encoder-only and encoder-decoder architectures have dominated prior ASR research, the potential of decoder-only transformer models for cross-lingual transfer learning remains relatively unexplored. Cross-lingual transfer learning aims to use knowledge from a high-resource language (e.g., English) to improve ASR performance in a lower-resource language by exploiting shared linguistic structures. The aim of this project is to train a decoder-only transformer on English ASR data, fine-tune it on limited ASR data of another language, and analyse its cross-lingual transfer capabilities against baselines. By exploring the potential of decoder-only transformers for cross-lingual ASR, this project aims to contribute to the development of more flexible and effective speech recognition systems, particularly for languages with limited labelled data.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Amishi Gangwar)

Acknowledgements

I want to sincerely thank my supervisor, Hao Tang, for his outstanding guidance throughout this research. His patience in answering my questions and suggesting approaches has been invaluable. This work would not have been possible without his support.

I'm also very grateful to my family and friends who encouraged me and stood by me during this journey.

Table of Contents

1	Introduction	1
1.1	Motivation	2
1.2	Research Question	2
1.3	Structure	3
2	Background and Relevant Work	4
2.1	Speech Representation	4
2.2	Transformers	5
2.2.1	Encoder-Decoder Transformers	5
2.2.2	Encoder-Only Transformers	6
2.2.3	Decoder-Only Transformers	8
2.3	Cross Lingual Transfer Learning in ASR	9
2.4	Criticism of Previous Work	10
3	Methodology	11
3.1	Model Architecture	11
3.1.1	Feature Extraction with HuBERT	11
3.1.2	Decoder-Only Transformer Architecture	12
3.2	Training Setup	14
3.2.1	Pre-training on English ASR	18
3.2.2	Fine-Tuning on Spanish ASR	19
4	Experimental Setup	20
4.1	Hypothesis	20
4.2	Datasets	20
4.3	Baseline	21
4.4	Experiments on English Pre-Training	22
4.5	Experiments on Spanish Fine-Tuning	22
4.6	Evaluation Metrics	22
5	Results and Analysis	24
5.1	English Pre-Training Results	24
5.1.1	Effect of Training Data on ASR Performance	24
5.1.2	Hyperparameter Optimization	25
5.1.3	Effect of Model Depth	25
5.2	Spanish Fine-Tuning Results	26

5.2.1	Effect of Data on Spanish Fine-Tuning	27
5.2.2	Baseline Comparison	28
5.3	Discussion of Findings	29
6	Conclusions	30
6.1	Summary of Contributions	30
6.2	Limitations	31
6.3	Future Work	31
	Bibliography	32

Chapter 1

Introduction

Human communication is predominantly speech-driven, making spoken language one of the most efficient means of information exchange. Automatic Speech Recognition (ASR), the technology that enables machines to understand and transcribe human speech, has been one of the most challenging and important goals in artificial intelligence research. The origins of the field can be traced back to early achievements such as the Digit Recognizer developed by Davis et al. (1961), which could identify spoken numerical digits by detecting unique patterns in audio. In recent years, ASR research has seen remarkable achievements with the rise of deep learning, particularly transformer-based architectures.

Traditional ASR models have largely relied on Hidden Markov Models (HMMs) and Recurrent Neural Networks (RNNs). However, transformer architectures have recently outperformed these methods due to their ability to model long-range dependencies and effectively capture complex speech representations. Modern ASR systems typically use encoder-only architectures, such as wav2vec 2.0 (Baevski et al., 2020), or encoder-decoder architectures like OpenAI’s Whisper (Radford et al., 2023). However, the rapid development of Large Language Models (LLMs), most prominently decoder-only transformers like the Generative Pre-Trained Transformer (GPT), has inspired an interest in integrating decoder-only models within ASR tasks. Such decoder-only architectures offer potential advantages, including architectural simplicity and unified language modelling capabilities (Wang et al., 2022), yet their effectiveness for ASR in cross-lingual transfer scenarios, remains underexplored.

Cross-lingual transfer learning presents a significant challenge in ASR, as it seeks to make use of high-resource languages (e.g., English) to develop effective models for languages with limited labelled data. This approach exploits shared linguistic structures and semantic similarities across languages to enhance performance in low-resource settings. Pre-trained or jointly trained encoder-decoder models do not share the language modelling component (decoder) for the same language, limiting their adaptability in multilingual settings (Wang et al., 2020). In contrast, a decoder-only approach could provide a more unified and adaptable framework for multilingual ASR.

This dissertation investigates the potential of decoder-only transformer architectures for

cross-lingual transfer learning in ASR. The aim is to train a decoder-only transformer on English ASR data, fine-tune it on limited ASR data of another language, and evaluate its transfer capabilities. Through comparative analyses against standard baseline models, this study aims to determine whether decoder-only architectures can achieve competitive cross-lingual performance. Insights from this project are expected to enhance the understanding of the adaptability and suitability of decoder-only transformer architectures and their contribution towards developing more robust and accessible ASR systems for low-resource languages.

1.1 Motivation

The growing popularity of LLMs has sparked an increased interest in integrating them with speech processing, leading to the emergence of decoder-only transformer-based ASR systems. Unlike traditional ASR architectures, which handle audio processing (encoder) and language modelling (decoder) separately, decoder-only models accept input sequences where speech features and text are integrated within a single input stream. This integrated approach may present advantages by simplifying the model architecture for ASR systems towards enhancing their adaptability for cross-lingual knowledge transfer.

Cross-lingual transfer learning has become increasingly important for ASR development as the digital language divide continues to widen. While high-resource languages like English benefit from abundant training data, thousands of languages worldwide remain excluded from speech technology due to limited resources. Transfer learning from high-resource languages is known to be an efficient way to improve end-to-end ASR for low-resource languages (Wang et al., 2020), significantly reducing the resources required to develop functional speech recognition across a broader linguistic spectrum. Improved methods for low-resource ASR, such as the decoder-only transformer, can enable applications such as dictation systems, voice assistants, and transcription services for languages with limited data resources.

1.2 Research Question

This dissertation seeks to address the following central research question:

Can a decoder-only transformer architecture, pre-trained on English ASR data and fine-tuned on limited data from a low-resource language, achieve competitive performance in cross-lingual transfer learning compared to baselines?

This question explores whether decoder-only transformers can effectively perform cross-lingual transfer learning in ASR, particularly examining how they might use data from high-resource languages to facilitate ASR for languages with limited resources. By exploring this question, the project aims to uncover whether the simplified architecture and unified language modelling of decoder-only transformers offer a viable alternative to traditional approaches in multilingual ASR. Figure 1.1 provides a more in-depth overview of the research methodology.

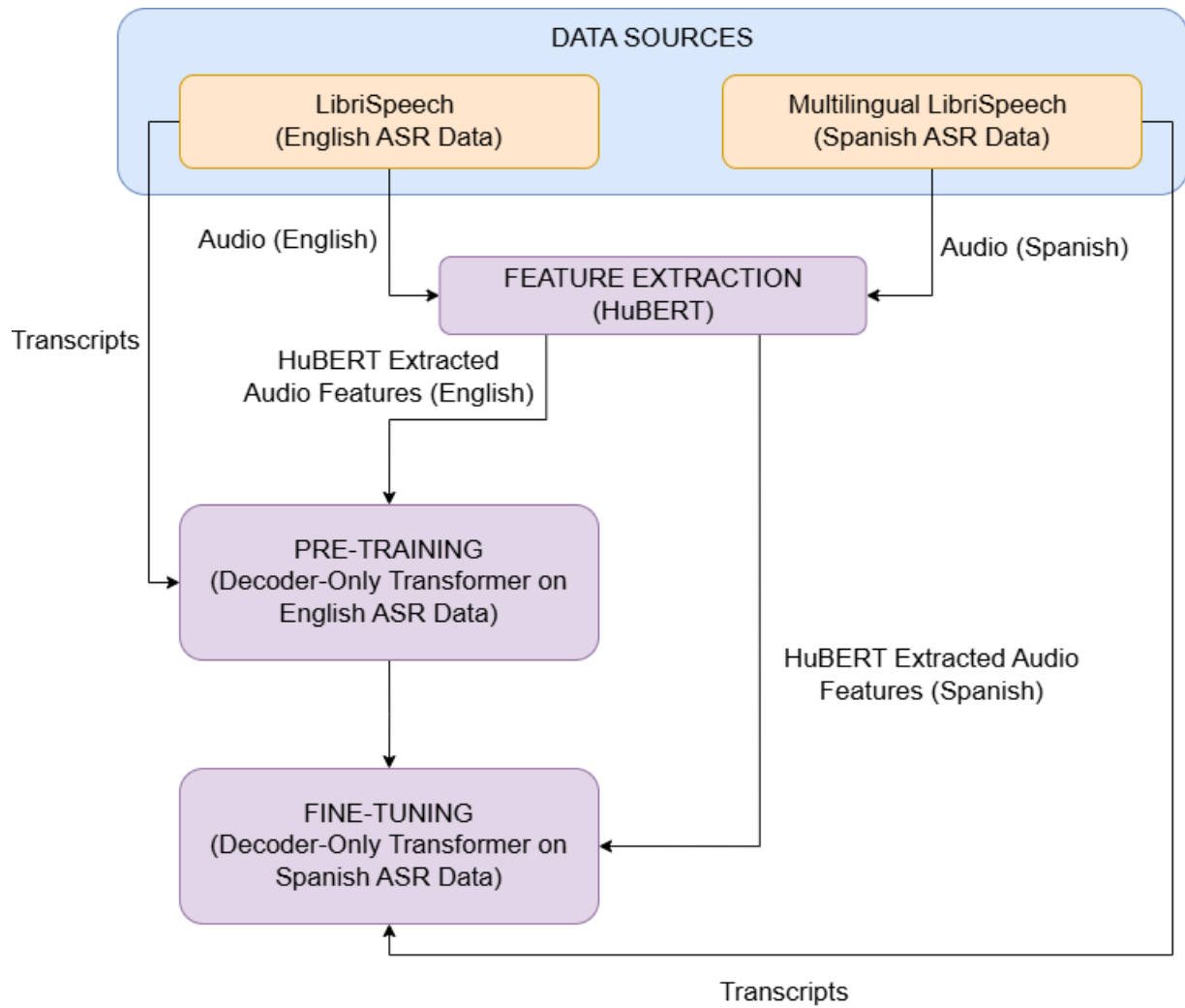


Figure 1.1: Overview of the Research Methodology

1.3 Structure

The remainder of this dissertation is structured as follows. Chapter 2 presents essential background information on automatic speech recognition, transformer architectures, and cross-lingual transfer learning, along with a critical review of relevant prior work on decoder-only models and low-resource ASR. Chapter 3 outlines the proposed methodology, detailing the decoder-only architecture, the use of HuBERT (Hsu et al., 2021) for feature extraction, and procedures for pre-training and fine-tuning. Chapter 4 describes the experimental setup, outlining the hypotheses, the datasets used, baseline comparison models, and evaluation metrics. Chapter 5 reports the experimental results, including detailed analyses of the performance of cross-lingual transfer compared to the established baseline models. Finally, Chapter 6 summarizes the key findings, highlights the contributions made by this project, acknowledges limitations, and proposes directions for future research.

Chapter 2

Background and Relevant Work

This chapter presents the foundational theoretical background and context needed to understand ASR systems. It starts with an overview of speech representation and then delves into transformer-based architectures for ASR. The discussion then shifts to cross-lingual transfer learning, exploring its critical role in adapting to low-resource languages. Lastly, the chapter surveys recent developments in decoder-only ASR models and cross-lingual transfer, pinpointing key limitations and research gaps that this dissertation seeks to address.

2.1 Speech Representation

ASR systems take audio as input and process it to generate a text transcription. Speech originates as pressure waves in the air, created by vibrations such as those of our vocal cords. To process these analogue signals digitally, computers sample the waveform at regular intervals, measuring the amplitude at each point. According to the Nyquist-Shannon sampling theorem (Por et al., 2019), a signal must be sampled at a rate at least twice the highest frequency present to accurately reconstruct the original sound without distortion or loss of information. Since human speech typically contains frequencies up to 8 kHz, ASR systems commonly use a 16 kHz sampling rate to ensure a balance between accuracy and computational efficiency. The raw audio waveform is the one-dimensional sequence of sampled amplitude values over time, representing the continuous variations in air pressure as a digital signal.

Recent developments have enabled modern transformer-based ASR models to process raw waveforms directly. These models first normalize the waveform and encode it into a sequence of feature vectors, where each vector captures information about a short segment of the original waveform. These embeddings are then processed by transformer layers, where self-attention mechanisms capture long-range dependencies and contextual relationships in the speech signal. By learning directly from raw audio, transformer-based ASR models achieve greater robustness to variations in accents, background noise, and speaking styles.

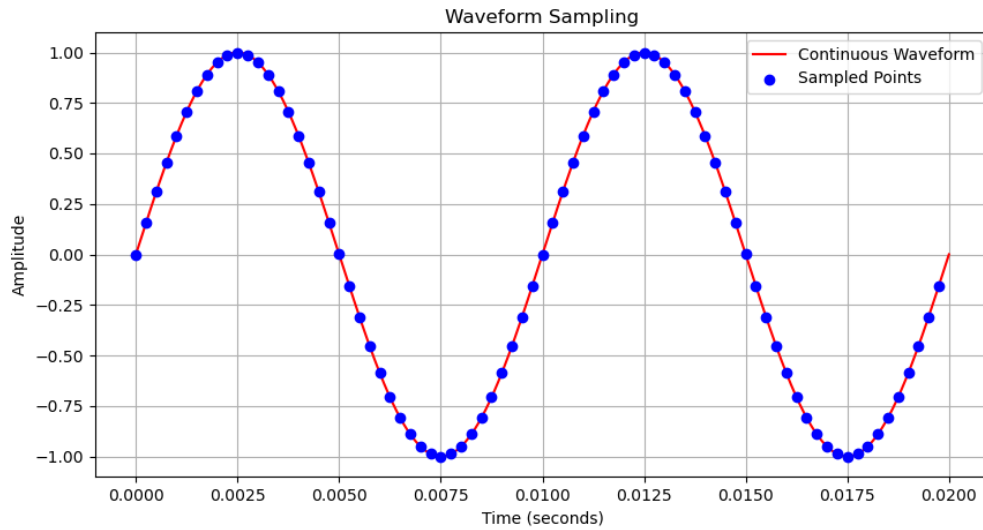


Figure 2.1: This figure demonstrates the process of converting a continuous speech waveform into a discrete digital representation through sampling. The red curve represents the original analog speech waveform, which varies continuously over time. To enable processing by ASR models, the signal is sampled at regular intervals, where each blue dot represents a discrete sample point.

2.2 Transformers

Transformers are deep learning architectures that process sequential data using self-attention mechanisms. Introduced in "Attention is All You Need" by Vaswani et al. (2017), these models capture long-range dependencies through direct weighted connections between all sequence elements. This parallel processing approach makes transformers highly scalable for sequential tasks such as ASR. This section examines three transformer variants: encoder-decoder, encoder-only, and decoder-only architectures.

2.2.1 Encoder-Decoder Transformers

Encoder-decoder transformers are typically designed for sequence-to-sequence tasks, including ASR, where an input speech signal is mapped to a corresponding text transcription. This architecture consists of two main components: the encoder and the decoder. In ASR systems, the encoder processes the raw audio waveforms to extract a contextualized representation of the audio. Given an input sequence of acoustic features, the encoder transforms it into a series of intermediate representations through multiple layers of self-attention and feedforward networks as seen in Figure 2.2. The decoder generates the corresponding transcription autoregressively, predicting one token at a time while attending to both previously generated tokens and the encoded speech representations. As seen in Figure 2.2, the decoder consists of self-attention layers, which allow the model to condition its predictions on prior tokens, and encoder-decoder attention layers, which enable the decoder to attend to the acoustic features encoded by

the encoder. To maintain causal generation, the self-attention mechanism in the decoder is masked, preventing the model from seeing future tokens in the output sequence during training.

Attention

Self-attention is the core operation in transformers and is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2.1)$$

where Q (queries), K (keys) and V (values) are learned representations of the input, d_k is the dimensionality of the key vectors and the softmax function converts a vector of K real numbers into a probability distribution of K possible outcomes to normalize the attention scores.

In ASR, the input is the raw audio waveform. The encoder first extracts a sequence of speech feature embeddings from the raw audio waveform, which is passed as input to the self-attention layers. This mechanism helps the model determine which parts of the speech sequence are most important for recognizing a given sound or word. Each speech frame (a short segment of audio, typically 25 ms long) is transformed into three different vectors: Query (Q) represents the current speech frame trying to find relevant information, Key (K) represents all speech frames that could provide relevant context and Value (V) contains the actual acoustic information of each frame, which is used to update the representation.

The multi-head attention mechanism, as seen in Figure 2.3 extends this by computing attention over multiple subspaces:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (2.2)$$

where each head computes the scaled dot-product attention independently, as follows:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (2.3)$$

2.2.2 Encoder-Only Transformers

Encoder-only transformer models consist solely of the encoder component from the original transformer architecture (Vaswani et al., 2017). Unlike encoder-decoder models, encoder-only models do not have a separate autoregressive decoder module to generate outputs token-by-token. Instead, these models typically produce representations that can be directly used for downstream tasks or combined with simpler linear layers for classification or sequence prediction tasks, such as ASR. This dissertation uses HuBERT (Hidden-Unit BERT) (Hsu et al., 2021) to extract features directly from raw waveforms.

HuBERT

HuBERT is a self-supervised speech representation learning model that employs a masked prediction objective inspired by BERT (Devlin et al., 2019). Self-supervised

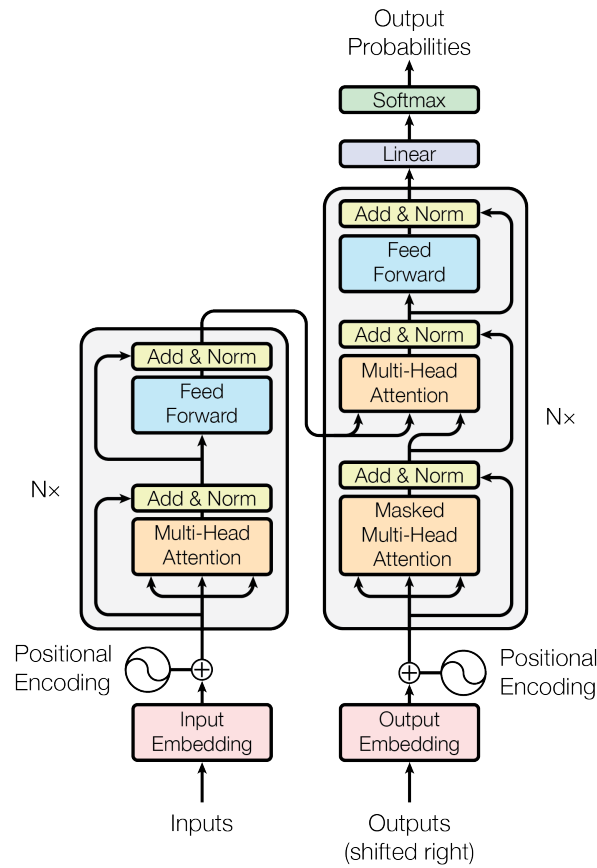


Figure 2.2: The transformer encoder-decoder architecture taken from Vaswani et al. (2017).

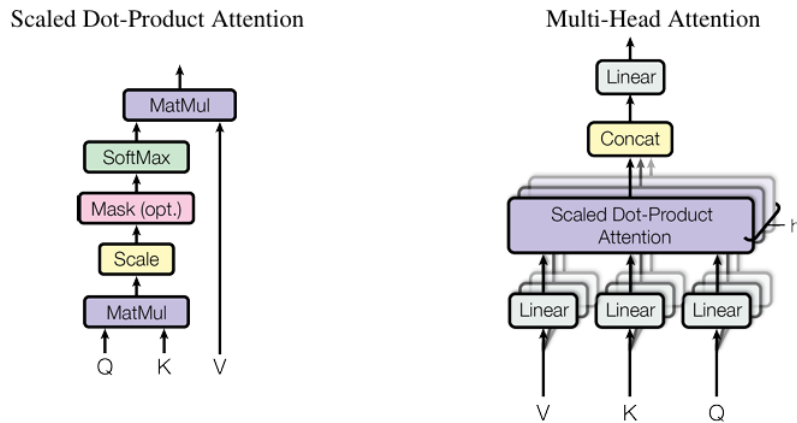


Figure 2.3: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel. Taken from Vaswani et al. (2017).

learning uses the intrinsic structure of unlabelled data to automatically generate pseudo-labels, enabling models to learn useful representations without needing manual annotations (Ericsson et al., 2022). HuBERT combines an offline clustering step with a BERT-like masked prediction task to learn speech representations. Its core idea is

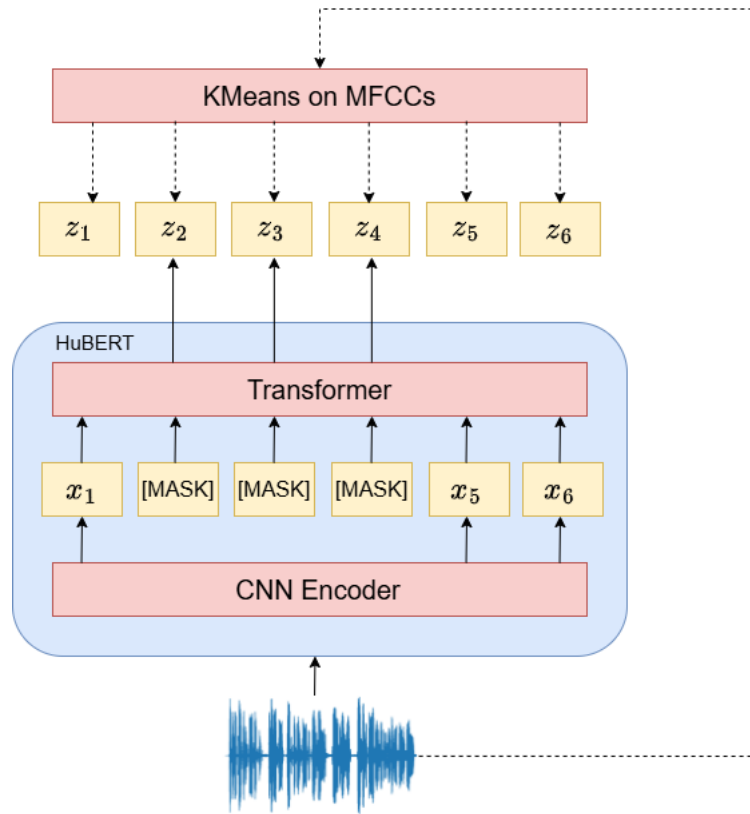


Figure 2.4: This figure illustrates HuBERT’s architecture and approach to extracting features. The input speech waveform is first processed into feature representations, which are then clustered using K-means to generate pseudo-labels (hidden units). During training, certain frames (x_2, x_3, x_4 in the figure) are masked, and the model learns to predict their hidden unit assignments based on the surrounding context. Diagram adapted from Hsu et al. (2021).

to predict hidden (latent) units derived from clustering continuous speech features, focusing the prediction loss only on masked regions of the input. This forces the model to learn both acoustic and sequential (language-like) patterns from the audio. HuBERT is used as a powerful self-supervised model for learning speech representations from unlabelled audio, which are then fine-tuned for ASR tasks.

2.2.3 Decoder-Only Transformers

The rise of large language models (LLMs) such as the Generative Pre-Trained Transformer (GPT) (Radford et al., 2018) has brought attention to decoder-only transformers, which omit the encoder and process both input and output sequences using a single model. While traditionally used in text generation tasks, decoder-only architectures are being investigated for ASR due to increasing interest in integrating LLMs with speech. In a decoder-only ASR system, the input consists of both the audio and text streams, concatenated into a single sequence (Gupta et al., 2024). The model then decodes this combined input into the desired transcription. In decoder-only transformers, the attention is unidirectional, where each position can only attend to itself and

previous positions through causal masking. The decoder-only setup is unconventional since the audio and text modalities are fundamentally different – audio is continuous and high-dimensional, while the text is discrete and symbolic. Despite this challenge, decoder-only models have shown promise in integrating multimodal data (Zhang et al., 2023), particularly in scenarios where large amounts of pre-training data are available. This flexibility not only simplifies ASR architectures but also suggests possibilities for broader applications where diverse data must be processed cohesively. One such avenue is cross-lingual transfer learning, where the focus shifts to utilizing pre-trained representations to address speech recognition across multiple languages. In this context, decoder-only transformers could potentially use their capacity to learn generalizable patterns from resource-rich languages and apply them to low-resource ones, tackling the challenge of linguistic diversity in ASR.

2.3 Cross Lingual Transfer Learning in ASR

Many ASR systems are optimized for only a small subset of the world’s approximately 7,000 spoken languages, leaving speakers of less common languages without access to speech-enabled technologies. The lack of transcribed speech data for these languages presents a significant challenge in developing effective ASR models. Recent developments in machine learning and transformers have enabled models to learn from languages with abundant resources and generalize to low-resource languages through cross-lingual transfer learning (Minixhofer et al., 2021). This approach uses knowledge from high-resource languages to improve recognition performance where labelled data is scarce.

Despite its promise, cross-lingual transfer learning in ASR faces several challenges. Phonetic variability between languages can lead to poor adaptation if the target language has phonemes not present in the source language. Additionally, morphological complexity varies significantly across languages; for instance, some languages have a richer inflectional system than English, making direct transcription transfer difficult (Ahmad et al., 2018). Acoustic differences such as intonation, stress, and rhythm further complicate adaptation, as models trained on one language’s speech patterns may struggle with another’s. Finally, writing system differences introduce an additional challenge, as some languages exhibit weak speech-text alignment due to distinct spoken and written forms. However, a comparison between encoder-only models and encoder-decoder models for building ASR systems showed that pre-trained models exposed to more data from a specific language family generally perform better on new, related languages within that family (Rouditchenko et al., 2023).

Although decoder-only cross-lingual transfer learning has been explored in large language models (LLMs) (Aggarwal et al., 2024), its application to ASR remains an open research question. Decoder-only ASR for multilingual transfer is still underexplored, making it unclear how well these models generalize across languages in speech tasks. This dissertation explores the effectiveness of decoder-only transformers in cross-lingual ASR by pre-training a model on English speech data and fine-tuning it on limited data of another language, evaluating its ability to generalize across linguistic differences. Spanish was chosen as a case study because, despite its phonetic differences and mor-

phonologically richer structure compared to English, it shares a largely similar character set with only a few additional accented letters. This allows for a controlled evaluation of how well a model pre-trained on English can adapt to a related yet distinct language.

2.4 Criticism of Previous Work

Research on decoder-only transformers for ASR is still in its early stages. Recent developments in ASR have explored decoder-only transformer architectures as a viable alternative to traditional encoder-decoder models. These models have gained attention due to their ability to unify speech and text processing, potentially simplifying ASR pipelines.

Decoder-Only Transformers: Tsunoo et al. (2023) proposed a decoder-only ASR model that uses Connectionist Temporal Classification (CTC) prompts to align speech with text and text data augmentation for robustness. Their approach reduced word error rates (WER) by 0.3% and 1.4% on LibriSpeech test-clean and test-other datasets, respectively, and by 2.9% and 5.0% on Switchboard and CallHome datasets. Similarly, Gupta et al. (2024) trained decoder-only models on public ASR datasets, demonstrating competitive performance by surpassing encoder-decoder baselines on 7 out of 15 English ASR benchmarks. These studies demonstrate that while decoder-only models hold promise, they have yet to be thoroughly investigated for cross-lingual transfer in ASR.

Cross Lingual Transfer Learning: Several studies have tackled cross-lingual ASR by employing different strategies. However, most existing approaches rely on encoder-decoder architectures. Wang et al. (2020) proposed incorporating speech-to-text translation as an auxiliary task to facilitate cross-lingual transfer. Their Speech Translation Enhanced Transfer model demonstrated up to 24.6% WER reduction compared to direct transfer from high-resource ASR. More recently, Pillai et al. (2024) proposed a multi-stage fine-tuning approach using intermediate models trained on linguistically similar languages. While effective, this study experiments with OpenAI’s Whisper model Radford et al. (2023), which is a large model trained on 680,000 hours of multilingual data in almost 100 languages and hence has better generalization capabilities.

Despite recent progress in decoder-only ASR and cross-lingual transfer learning, existing research has not fully explored decoder-only architectures in multilingual ASR settings. Most studies on decoder-only transformers focus on monolingual ASR, evaluating performance improvements on English datasets while leaving their cross-lingual adaptability largely untested. Conversely, research on cross-lingual ASR has primarily centered around encoder-decoder architectures, with little attention given to how decoder-only models transfer across languages. This dissertation aims to bridge this gap by investigating whether a decoder-only transformer trained on English ASR data can effectively generalize to Spanish through direct fine-tuning, thereby evaluating the potential of decoder-only architectures for cross-lingual ASR.

Chapter 3

Methodology

This chapter presents the methodology for training and evaluating a decoder-only transformer for cross-lingual ASR. It begins by describing the model architecture, including how the pre-trained HuBERT model is used for feature extraction and how the decoder processes speech representations to generate transcriptions. The implementation details of the decoder architecture are then outlined, followed by a discussion of the training setup, which consists of pre-training on English speech data and fine-tuning on Spanish to assess cross-lingual transferability.

3.1 Model Architecture

The implemented model follows a decoder-only transformer architecture that relies on a pre-trained HuBERT model for feature extraction. The model uses HuBERT to process raw speech waveforms into contextualized speech embeddings. These extracted features serve as input to the decoder, which autoregressively generates transcriptions. Unlike traditional encoder-decoder ASR models, where the encoder dynamically learns acoustic representations, in this approach, the encoder (HuBERT) remains frozen during training, ensuring that all learnable parameters reside within the decoder.

3.1.1 Feature Extraction with HuBERT

The model utilizes a pre-trained HuBERT Base encoder, which has been trained on 960 hours of LibriSpeech data (Ott et al., 2019). HuBERT Base consists of 12 transformer encoder layers, each with 768 hidden dimensions, 8 attention heads, and a feedforward layer of size 3,072. It has 95 million parameters and operates on 16 kHz audio, generating frame-wise hidden state representations, with each frame corresponding to approximately 20 ms of speech (Hsu et al., 2021). During pre-training, HuBERT is optimized using a masked prediction objective, allowing it to learn robust speech representations without requiring transcribed data. In this implementation, the HuBERT encoder remains frozen, ensuring that only the decoder is trained for ASR.

Given an input waveform x , the HuBERT encoder processes it through multiple transformer layers, producing hidden state representations:

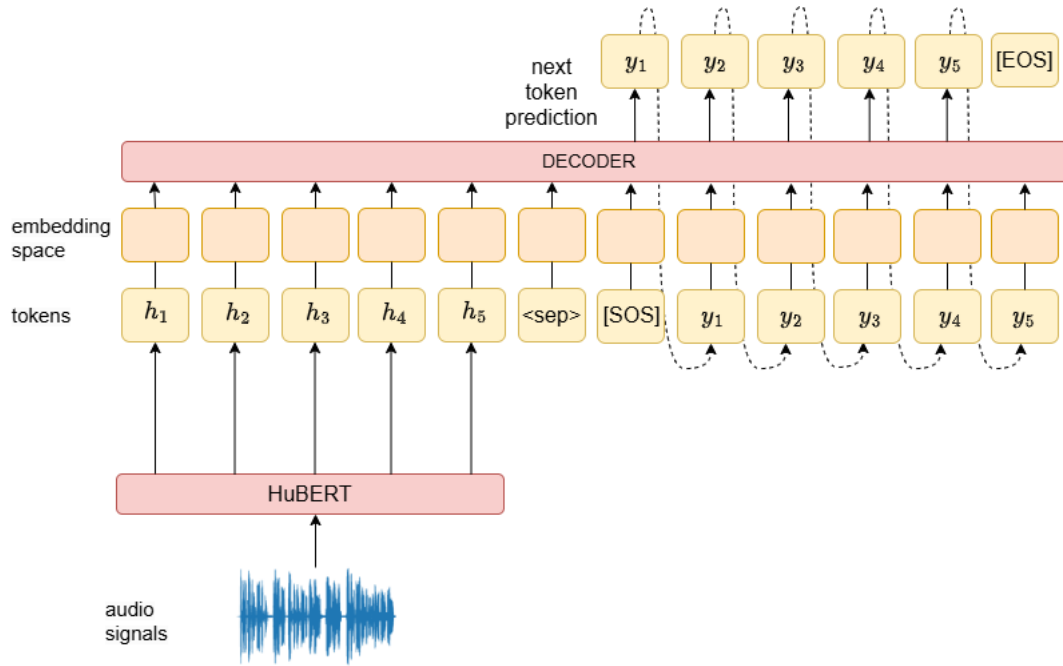


Figure 3.1: Model architecture of the decoder-only transformer for ASR.

$$h = \text{HuBERT}(x) \quad (3.1)$$

where h is a sequence of contextualized speech embeddings. These embeddings are passed directly to the decoder.

3.1.2 Decoder-Only Transformer Architecture

The implementation of the decoder-only transformer follows the methodology outlined in Starmer (2024) and is designed to match the dimensions of the HuBERT Base Model. Specifically, the decoder consists of 12 transformer layers, aligning with the 12-layer architecture of the HuBERT Base model used for feature extraction. The decoder consists of multi-head self-attention layers, feedforward network layers, and layer normalization.

Attention and Causal Masking

Each layer of the decoder applies multi-head self-attention, as shown in Equation 2.2, where each token attends only to previous tokens due to causal masking. The attention mechanism is computed as shown in Equation 2.1. The attention outputs are computed across multiple heads, allowing the model to capture different contextual relationships in the input sequence.

Feedforward Networks and Layer Normalization

Each self-attention layer is followed by a position-wise feedforward network (FFN) that consists of two fully connected layers with a Rectified Linear Unit (ReLU) activation (Vaswani et al., 2017):

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (3.2)$$

where x is the input vector of embeddings, and W_1 and W_2 are learnable weight matrices. The first linear transformation projects the input into a higher-dimensional space, here four times the model's hidden size. The ReLU activation function (Nair and Hinton, 2010) is defined as:

$$\text{ReLU} = \max(0, x) \quad (3.3)$$

which means it outputs zero for negative values and retains positive values unchanged. This introduces non-linearity, preventing the network from collapsing into a purely linear function. By allowing only positive activations to propagate, ReLU helps in learning complex patterns while mitigating issues like vanishing gradients (Nair and Hinton, 2010). The second linear transformation projects the representation back to the original hidden size, preserving the model's capacity. This structure enables the model to learn complex mappings beyond what is possible with self-attention alone.

Layer Normalisation

To improve training stability, layer normalization is applied after each self-attention and feedforward network operation (Ba et al., 2016):

$$\text{LayerNorm}(x) = \frac{x - \mu}{\sigma + \epsilon} \quad (3.4)$$

where μ and σ are the mean and variance of the input activations.

Decoder Input Representation

Unlike encoder-decoder architectures, the decoder-only model processes both the extracted speech representations and tokenized text in a unified sequence (Tsunoo et al., 2023). As seen in Figure 3.2, for each training step, the decoder takes as input:

- Extracted features h from the HuBERT feature extractor, which provides contextualized speech representations.
- A separator embedding s , a learned vector that marks the boundary between speech and text.
- Target token embeddings w , which correspond to the ground truth transcription sequence.

These components are concatenated to form the decoder input sequence:

$$[h, s, w] \quad (3.5)$$

Positional Encodings

Since transformers lack recurrence, positional encodings are added to text embeddings to retain word order information. The positional encodings are computed as (Vaswani et al., 2017):

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right), \quad PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right) \quad (3.6)$$

where pos is the position index and d is the embedding dimension. These encodings are added to the text token embeddings (w) before passing through the attention mechanism. This ensures that the model can differentiate token positions and understand word order.

Text Token Embeddings

A learned embedding layer maps text tokens to dense vectors, which are processed by the self-attention layers:

$$e_w = \text{Embedding}(w) + PE \quad (3.7)$$

3.2 Training Setup

Training an ASR model involves several key components, including defining the vocabulary, data pre-processing, and model optimization. This section outlines the setup used for training the decoder-only ASR model, detailing the vocabulary structure, data handling, and model configurations.

Vocabulary

In any ASR system, the model must convert spoken language into written text, which requires a well-defined vocabulary to map speech features to textual representations. The vocabulary serves as the bridge between audio input and textual output, ensuring that the model can correctly transcribe speech into a structured sequence of tokens.

For this model, the vocabulary is defined and loaded from a JSON file, which contains a mapping of text tokens to their corresponding indices. This vocabulary is used to tokenize text sequences during training and inference. The model employs a character-level vocabulary, meaning each letter, punctuation mark, and special token (e.g., start-of-sequence and end-of-sequence tokens) has a unique ID, as seen in Table 3.1. The only punctuation mark included in the vocabulary is the apostrophe ('), as it is essential for correctly transcribing contractions (e.g., *don't*, *it's*) and possessive forms (e.g., *student's*). Other punctuation marks are excluded to simplify the decoding process and focus on the core phonetic transcription task.

The vocabulary is initialized using a tokenizer. This tokenizer enables character-based tokenization with the following special tokens:

- [SOS] (Start of Sequence) – Marks the beginning of a transcription.
- [EOS] (End of Sequence) – Signals that transcription has ended.
- [PAD] (Padding Token) – Used to pad sequences to a fixed length for batch processing.
- [UNK] (Unknown Token) – Represents out-of-vocabulary (OOV) characters.

Character	Index	Character	Index	Character	Index	Character	Index
a	0	i	8	q	16	y	24
b	1	j	9	r	17	z	25
c	2	k	10	s	18	” ” (space)	26
d	3	l	11	t	19	[PAD]	27
e	4	m	12	u	20	[UNK]	28
f	5	n	13	v	21	,	29
g	6	o	14	w	22	[SOS]	30
h	7	p	15	x	23	[EOS]	31

Table 3.1: Character-Level Vocabulary Mapping

During training, text sequences are tokenized into indices before being fed into the model. An example of tokenization can be seen in Table 3.2.

[SOS]	h	e	l	l	o	” ”	w	o	r	l	d	[EOS]
30	7	4	11	11	14	26	22	14	17	11	3	31

Table 3.2: Example of Tokenization and Index Mapping

Dataset Processing

With the vocabulary defined, the next step involved preparing the training and evaluation datasets. This included extracting features from raw speech waveforms, tokenizing transcriptions, and organizing data into structured batches suitable for training the model.

Each audio file is loaded and processed using `torchaudio`. The waveform is used in its original sampling rate (16 kHz), and supplied to HuBERT. Since HuBERT operates directly on raw waveforms, no handcrafted feature extraction is needed.

The corresponding transcriptions are first converted to lowercase to ensure consistency. Each text sequence is then appended with the special tokens `[SOS]` and `[EOS]` to indicate sentence boundaries. To handle varying sequence lengths within a batch, shorter sequences are padded with the `[PAD]` token, ensuring uniform input sizes for efficient training.

Once the dataset is preprocessed and structured for training, the next step involves optimizing the model’s learning strategy. A technique used during training is teacher forcing (Williams and Zipser, 1989), which helps improve convergence and reduce exposure bias.

Teacher Forcing

During training, the model employs teacher forcing, where the ground-truth tokens from the dataset are provided as inputs to guide the learning process. The decoder predicts the next token $\hat{y}_t$, but instead of using its own previously generated token $\hat{y}_{t-1}$, it is fed the actual ground-truth token y_{t-1} as input. Teacher forcing is helpful because it accelerates training by ensuring the model learns from correct sequences rather than compounding errors from early mistakes. It also reduces exposure bias, which occurs

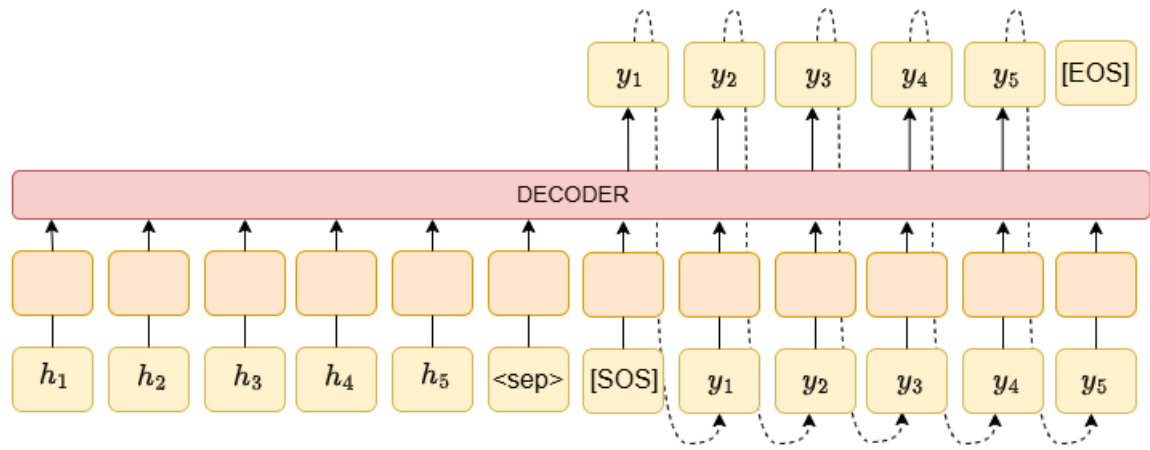


Figure 3.2: Input to the decoder, where $h_1, \dots, h_5$ represent the extracted speech representations, sep represents the separator embedding and $y_1, \dots, y_5$ represent the ground truth embedding vectors.

when a model is trained with correct previous tokens but must generate tokens in an autoregressive manner, during inference. In ASR, exposure bias can lead to cascading errors, where an early mistake distorts the entire transcription.

During training, where teacher forcing ensures stable learning, the model is optimized using a loss function that quantifies the discrepancy between its predictions and the actual ground truth tokens.

Loss Function

The model is trained using the cross-entropy loss, which measures the difference between the predicted probabilities and the ground-truth transcription sequence. Cross-entropy loss is commonly used for training Transformer models (Niu et al., 2024).

The cross-entropy loss is computed against the ground-truth transcription sequence as follows:

$$\mathcal{L} = - \sum_t y_t \log \hat{y}_t \quad (3.8)$$

where y_t is the true token and $\hat{y}_t$ is the predicted probability from the softmax output. To avoid unnecessary penalties on padding, the loss function applies an ignore mask to the [PAD] token.

Once training is complete, the model is evaluated in inference mode, where it no longer has access to ground truth tokens. Instead, it generates text in an autoregressive manner, as described below.

Autoregressive Token Generation

The decoder-only transformer generates text sequentially, producing one token at a time in an autoregressive manner. This means that at each time step t , the model predicts the next token y_t based on the previously generated tokens $y_{<t}$, i.e., all tokens before time step t , and the extracted HuBERT features h .

Mathematically, this process is defined as:

$$P(y_t|y_{<t}, h) = \text{softmax}(W_{\text{out}} \cdot x_t) \quad (3.9)$$

where x_t is the hidden state produced by the transformer at time step t , W_{out} is a learnable projection matrix that maps the hidden state into a logit vector of size equal to the vocabulary. Logits are raw, unnormalized scores produced by the model before applying the softmax function. They indicate the model's confidence in each token. The softmax function converts the logits into a probability distribution over all possible tokens, ensuring that the model assigns a probability to each word in the vocabulary.

The token generation process begins with a Start-of-Sentence [SOS] token, which signals the model to start decoding. The decoder then operates in an autoregressive manner, predicting the next token based on the previously generated tokens and the extracted HuBERT features. Once a token is predicted, it is appended to the input sequence, allowing the model to iteratively refine its predictions. This process continues until the model outputs an End-of-Sentence [EOS] token, marking the completion of the transcription.

The selection of each predicted token is determined by the following equation:

$$\hat{y} = \text{argmax} P(y_t|y_{<t}, h) \quad (3.10)$$

Here, $\hat{y}$ represents the predicted token at time step t , chosen as the token with the highest probability from the distribution. $P(y_t|y_{<t}, h)$. The argmax operation identifies the index of the maximum value in the probability distribution, effectively selecting the most likely token from the vocabulary based on the model's current understanding of the sequence and audio context. This greedy decoding strategy ensures that, at each step, the model commits to the token it deems most probable, building the transcription incrementally until completion.

Hyperparameter Selection

The next step in the training setup involved selecting appropriate hyperparameters to ensure stable training of the model. Key hyperparameters were carefully chosen based on prior work in transformer-based ASR models and experimental tuning.

- **Model Hidden Size:** The decoder uses a hidden size of 768, matching the HuBERT base model for compatibility with extracted speech representations.
- **Number of Layers:** The decoder consists of 12 transformer layers, mirroring HuBERT's depth to maintain balanced representation learning.
- **Attention Heads:** Each self-attention layer employs 8 attention heads, matching the configuration used in HuBERT.
- **Dropout Rate:** A dropout of 0.1 (Vaswani et al., 2017) is applied after self-attention and feedforward layers to prevent overfitting (Escolano et al., 2024). Dropout is a regularization technique, that functions by randomly deactivating

(setting to zero) certain connections within the network during the training phase. This approach compels the network to adapt to functioning with a reduced number of units, thereby enhancing its robustness against overfitting (Li and Ke, 2024).

- **Batch Size:** The batch size is set to 4, constrained by GPU memory due to long speech sequences. Batching allows the model to process multiple examples in parallel, making training more efficient.
- **Learning Rate:** The Adam optimizer is used with a learning rate of $1e-4$, selected after comparing multiple values, as explained in Chapter 5.
- **Positional Encoding Limit:** The model supports a maximum sequence length of 1024 tokens, ensuring robustness for long transcriptions.

3.2.1 Pre-training on English ASR

The initial phase of training involves pre-training the decoder-only transformer on English speech data. Since English is a high-resource language with abundant annotated speech data, pre-training on English enables the model to learn fundamental acoustic and linguistic patterns before adapting to a lower-resource language like Spanish.

The pre-training process begins with loading and splitting the dataset into training (70%), validation (15%), and test (15%) subsets. The training set optimizes the model's parameters by exposing it to a substantial portion of the data, allowing the learning of English speech-to-text patterns, the validation set monitors performance on unseen data to tune hyperparameters and prevent overfitting and the test set, reserved for final evaluation, provides an unbiased measure of the pre-trained model's generalization ability.

During training, an utterance is read from the dataset as a waveform, along with its sampling rate (16KHz), and ground-truth transcript (e.g., "hello"). The utterance is batched with $x - 1$ others, (if the batch size is x) via the `collate_fn`. In the `collate_fn`, the waveform is padded to match the batch's longest sequence, and the transcript is modified to "[SOS] hello [EOS]" before being tokenized into indices (e.g., [30, 7, 4, 11, 11, 14, 31]) using the character-level vocabulary shown in Table 3.1.

The batch's waveforms are then processed by HuBERT and the raw audio is converted into contextualized features. These are specifically extracted from the ninth hidden layer of HuBERT and represent the audio prefix. Vielzeuf (2024) identified the ninth layer of the HuBERT base model as the best layer in terms of representation quality, indicating that this layer captures high-level linguistic information more effectively compared to other layers. The tokenized transcript (excluding [EOS], e.g., [30, 7, 4, 11, 11, 14]) is embedded into 768-dimensional vectors, augmented with positional encodings to preserve sequence order.

A learned separator embedding is appended to the audio features, followed by the embedded text, forming a unified sequence. This embedding allows the decoder to differentiate between the audio features and the embedded text. This sequence passes through the transformer layers, where multi-head self-attention with causal masking and feedforward networks compute dependencies and refine representations, producing

logits over the vocabulary for the text portion. The logits are aligned with the shifted ground-truth tokens (e.g., [7, 4, 11, 11, 14, 31]) to compute cross-entropy loss.

The optimizer updates the model’s parameters based on the computed loss, adjusting the weights to minimize the prediction error. Training is conducted for a predefined number of epochs, with the loss monitored to check if the validation loss stagnates or increases, preventing overfitting.

Once pre-training is complete, the final model is evaluated on the test set, providing an unbiased assessment of its performance. The learned parameters serve as an initialization for fine-tuning on Spanish, using the knowledge acquired from English ASR to improve transcription accuracy in the lower-resource language.

3.2.2 Fine-Tuning on Spanish ASR

After pre-training the decoder-only transformer on English ASR, the model is fine-tuned on Spanish speech data to adapt to the phonetic and linguistic differences of the target language. The fine-tuning phase refines the model’s learned representations while using knowledge from English to improve transcription performance in a lower-resource setting.

Fine-tuning follows the same data processing pipeline as pre-training, where Spanish speech waveforms are paired with ground-truth transcripts and tokenized into character-level sequences. The dataset is split into training, validation, and test subsets, ensuring that performance is tracked throughout training. The HuBERT encoder, frozen during both pre-training and fine-tuning, extracts high-level audio representations from the 9th hidden layer, maintaining consistency in feature extraction. Zhao and Zhang (2022) showed that HuBERT learns robust speech representations that generalize well across languages, making it suitable for cross-lingual speech recognition. This allows HuBERT to extract meaningful features from Spanish speech, even though it was initially trained on English data.

Since Spanish phonetics differ from English, the decoder’s vocabulary is updated to include Spanish-specific characters as shown in Table 3.3, ensuring accurate tokenization. The model is then trained using cross-entropy loss on Spanish transcriptions, optimizing parameters to improve recognition accuracy in the target language.

Character	Index
á	32
é	33
í	34
ó	35
ú	36
ñ	37
ü	38

Table 3.3: Spanish-specific characters in the vocabulary added to the original English vocabulary shown in Table 3.1.

Chapter 4

Experimental Setup

4.1 Hypothesis

This dissertation investigates the effectiveness of a decoder-only transformer trained on English ASR and then fine-tuned on Spanish for cross-lingual transfer learning in speech recognition. Based on prior research in cross-lingual transfer learning in ASR, the following hypothesis is formulated:

Hypothesis: Cross-Lingual Transfer from English to Spanish achieves Competitive ASR Performance

A decoder-only transformer pre-trained on English ASR and fine-tuned on Spanish ASR is hypothesized to achieve strong transcription accuracy in Spanish despite being initially trained on English. The model is expected to generalize the phonetic and linguistic patterns learned from English, allowing an effective adaptation to Spanish. The evaluation metrics on Spanish are expected to be comparable to or better than existing models.

4.2 Datasets

To evaluate the hypothesis, high-quality speech datasets are required that provide sufficient linguistic diversity and transcription accuracy. This section details the datasets used to train and evaluate the models.

LibriSpeech (English ASR Dataset)

LibriSpeech is a widely used benchmark dataset for English ASR, derived from public domain audiobooks (Panayotov et al., 2015). It consists of approximately 1,000 hours of transcribed speech sampled at 16 kHz, making it one of the largest and most diverse English ASR datasets. LibriSpeech’s extensive speaker diversity, well-structured transcriptions, and large-scale availability make it ideal for training transformer models. Given its high-resource nature, it serves as the foundation for pre-training the decoder-only transformer model, allowing it to learn robust acoustic and linguistic representations before fine-tuning on Spanish ASR.

Other English ASR datasets, such as the Wall Street Journal (WSJ) dataset for ASR (Paul and Baker, 1992) and TED-LIUM (Rousseau et al., 2012), also offer high-quality transcriptions. However, WSJ is significantly smaller, making it less suitable for training transformer models requiring large-scale data. TED-LIUM provides transcribed TED talks, which contain more spontaneous speech patterns than audiobooks but lack the scale and structure of LibriSpeech. Given these considerations, LibriSpeech was chosen for its balance of dataset size, transcription quality, and speaker diversity.

Multilingual LibriSpeech (MLS) (Spanish ASR Dataset)

Multilingual LibriSpeech (MLS) is an extension of LibriSpeech designed for multilingual ASR research (Pratap et al., 2020). Like LibriSpeech, MLS is derived from audiobooks and maintains a 16 kHz sampling rate, ensuring consistency in audio quality across languages. For Spanish ASR, the dataset comprises approximately 1,500 hours of transcribed speech, making it one of the largest high-quality Spanish ASR datasets available. While other multilingual ASR datasets exist, such as Common Voice (Ardila et al., 2019), MLS was chosen due to its scale and transcription quality and also to ensure consistency between the pre-training and fine-tuning phases.

4.3 Baseline

To evaluate the performance of the decoder-only transformer model for fine-tuning, it is compared against the following baseline:

HuBERT fine-tuned on Spanish ASR data with CTC Loss

As a baseline, we select HuBERT fine-tuned on Spanish ASR using Connectionist Temporal Classification (CTC) loss, aligning with its original fine-tuning strategy (Hsu et al., 2021). As mentioned in Section 2.2.2, HuBERT is a self-supervised speech model that learns meaningful representations from large-scale unlabelled audio data, making it highly effective in multilingual and low-resource ASR settings (Zhao and Zhang, 2022). HuBERT’s masked prediction objective learns robust speech representations that capture phonetic information useful across languages. The baseline model consists of the HuBERT encoder pre-trained on the English LibriSpeech dataset and subsequently fine-tuned on Spanish ASR data, both using CTC loss.

CTC Loss is designed for tasks where alignment between sequences is difficult, e.g. aligning each character to its location in an audio file. CTC loss enables the model to learn flexible alignments between speech frames and text. Fine-tuning HuBERT on Spanish ASR data with CTC loss serves as a strong baseline because it employs an encoder-only architecture, allowing for a direct comparison with the decoder-only transformer model. Evaluating the decoder-only approach against HuBERT fine-tuned on Spanish ASR provides insights into whether decoder-only models can achieve competitive transcription accuracy in cross-lingual transfer learning.

4.4 Experiments on English Pre-Training

To evaluate the effectiveness of the decoder-only transformer model for English ASR, a series of experiments were conducted using the LibriSpeech dataset, with a 70% training, 15% validation, and 15% test split. These experiments aimed to evaluate the performance of the model under different training conditions as detailed below.

The first set of experiments focused on testing the feasibility of training a decoder-only transformer for ASR. To ensure the model was functioning correctly, training was initially conducted on progressively larger subsets of LibriSpeech—100 hours, 360 hours, and finally, the full 960-hour dataset. This allowed for an assessment of how increasing training data impacted the evaluation metrics. Once stable performance was achieved on the full 960-hour dataset, a series of learning rate experiments were conducted to determine their impact on convergence and accuracy.

Following this, the next phase involved evaluating the effect of model depth. The number of transformer layers was varied (6, 8, and 12 layers) to determine its influence on transcription accuracy. After confirming that a 12-layer decoder trained on the full dataset performed competitively, this configuration was used for Spanish fine-tuning.

4.5 Experiments on Spanish Fine-Tuning

To evaluate the performance of the decoder-only transformer model in cross-lingual transfer learning for ASR, the pre-trained English model was fine-tuned on Spanish ASR data from the Multilingual LibriSpeech (MLS) dataset. This fine-tuned model was compared against the HuBERT model fine-tuned on Spanish ASR data, as mentioned in Section 4.3. Additional experiments were performed to analyze how much Spanish ASR data is required for effective transfer learning. By gradually increasing the amount of available Spanish ASR data, the experiments aim to determine the minimum amount of data required to achieve good transcription accuracy.

4.6 Evaluation Metrics

To evaluate the performance of the decoder-only transformer, three key metrics are used: Character Error Rate (CER), Word Error Rate (WER) and Cross-Entropy Loss.

Character Error Rate (CER)

CER measures transcription accuracy at the character level and is calculated as:

$$\text{CER} = \frac{(S + I + D)}{N} \quad (4.1)$$

where S , I and D represent the total number of substitutions, insertions and deletions to obtain the ground truth, respectively and N is the total number of characters in the ground truth. This metric provides a fine-grained view of the model's ability to transcribe effectively.

This metric is particularly important as the vocabulary is character-level, as seen in Table 3.1, and the decoder generates output one character at a time. As a result, CER offers a fine-grained evaluation of the model’s transcription capabilities and is sensitive to minor character-level errors. An example calculation of CER is shown in Table 4.1

Reference Text (Ground Truth)	hello world
Hypothesis Text (Predicted)	hallo wurld
Substitutions (S)	2 (e → a, o → u)
Deletions (D)	0
Insertions (I)	0
Total Characters in Reference (N)	11
CER Calculation	$\frac{2+0+0}{11} = 0.1818$

Table 4.1: Example calculation of Character Error Rate (CER)

Word Error Rate (WER)

WER measures transcription accuracy at the word level and is calculated similarly to CER as:

$$\text{CER} = \frac{(S + I + D)}{N} \quad (4.2)$$

where S , I and D again represent the total number of substitutions, insertions and deletions to obtain the ground truth, respectively and N is the total number of words in the ground truth. WER provides a more holistic view of the model’s ability to produce coherent transcriptions. An example calculation of WER is shown in Table 4.2

Reference Sentence	I love natural language processing
Hypothesis Sentence	I like language process
Substitutions (S)	2 (love → like, processing → process)
Deletions (D)	1 (natural)
Insertions (I)	0
Total Words in Reference (N)	5
WER Calculation	$\frac{2+1+0}{5} = 0.6$

Table 4.2: Example calculation of Word Error Rate (WER)

Cross Entropy Loss

As shown in Equation 3.8, the cross-entropy loss quantifies the divergence between the model’s predicted probability distribution over the vocabulary and the true token sequence, with padding tokens masked to focus on meaningful predictions. Lower values indicate higher confidence and accuracy in token generation.

Chapter 5

Results and Analysis

This chapter presents an in-depth analysis of the experimental results for training a decoder-only transformer for ASR and evaluating its effectiveness in cross-lingual transfer learning. The goal is to determine whether the decoder-only architecture can achieve competitive performance in cross-lingual transfer learning in ASR by training the model on a high-resource language (English), followed by fine-tuning on the low-resource language (Spanish). The following sections present the results in detail, beginning with the experiments on English ASR training, before analyzing the impact of fine-tuning on Spanish ASR and finally closing with a discussion of findings.

5.1 English Pre-Training Results

All experiments evaluating the performance of the decoder-only transformer model were conducted on the `dev-clean` split of the LibriSpeech dataset to ensure consistency. The following subsections present results from experiments exploring the effect of training data size, model depth, and learning rate on transcription accuracy, using Character Error Rate (CER), Word Error Rate (WER) and Cross Entropy Loss as evaluation metrics.

5.1.1 Effect of Training Data on ASR Performance

To analyze the impact of training data size on ASR performance, the decoder-only transformer was trained on three subsets of the LibriSpeech dataset: 100 hours, 360 hours, and 960 hours. The performance of each model was evaluated using Cross Entropy Loss, Character Error Rate (CER) and Word Error Rate (WER). The results, summarized in Table 5.1, indicate that increasing the amount of training data leads to substantial improvements in transcription accuracy.

To further illustrate the impact of training data size on transcription quality, Table 5.2 presents example transcriptions generated by models trained on 100h, 360h, and 960h of data. The results demonstrate that models trained on smaller datasets (100h and 360h) produce more phonetic errors, misrecognizing "kaliko" as "cergo" and "calego," respectively. As the amount of training data increases to 960h, the model successfully

Training Data	CER ↓	WER ↓	Cross-Entropy Loss ↓
100h	28%	44%	0.3
360h	25%	40%	0.12
960h	8%	18%	0.09

Table 5.1: ASR performance on different amounts of training data.

transcribes the phrase without errors, highlighting the significant improvements in phonetic accuracy and word recognition that come with more training data.

Model	Predicted Transcription
Ground Truth	...kaliko hesitated...
100h Output	...cergo hesitated...
360h Output	...calego hesitated...
960h Output	...kaliko hesitated...

Table 5.2: Comparison of transcriptions across different training data sizes.

In general, this suggests that increasing the size of the training data significantly improves the performance of ASR.

5.1.2 Hyperparameter Optimization

Experiments were also conducted to refine training hyperparameters. The primary focus was on tuning the learning rate to enhance convergence. The model was trained using four different learning rates: 5×10^{-4} , 1×10^{-4} , 5×10^{-5} , and 1×10^{-5} . As shown in Figure 5.1, the choice of learning rate significantly impacts convergence speed and final ASR performance.

The highest learning rate (5×10^{-4} , red curve) led to rapid loss reduction within the first epoch, converging quickly to a low final loss. However, it plateaued early, suggesting that while it reached a reasonable loss quickly, it may have converged to a suboptimal local minimum. The learning rate of 1×10^{-4} (green curve) also showed fast convergence but maintained a more stable decline, achieving a comparable final loss to the highest learning rate.

For lower learning rates, 5×10^{-5} (orange curve) exhibited a steadier decline and converged to a similar final loss but exhibited slower convergence. The lowest learning rate (1×10^{-5} , blue curve) resulted in the slowest training process, with a significantly higher loss at each epoch and no clear convergence within the observed epochs. Overall, the results indicate that 1×10^{-4} provides the best balance between convergence speed and final loss, making it the most effective learning rate choice.

5.1.3 Effect of Model Depth

To investigate the effect of model depth on ASR performance, the decoder-only transformer was trained with varying layers: 6, 8, and 12. All models were trained on the full 960-hour LibriSpeech dataset to ensure a fair comparison. The goal of this experiment

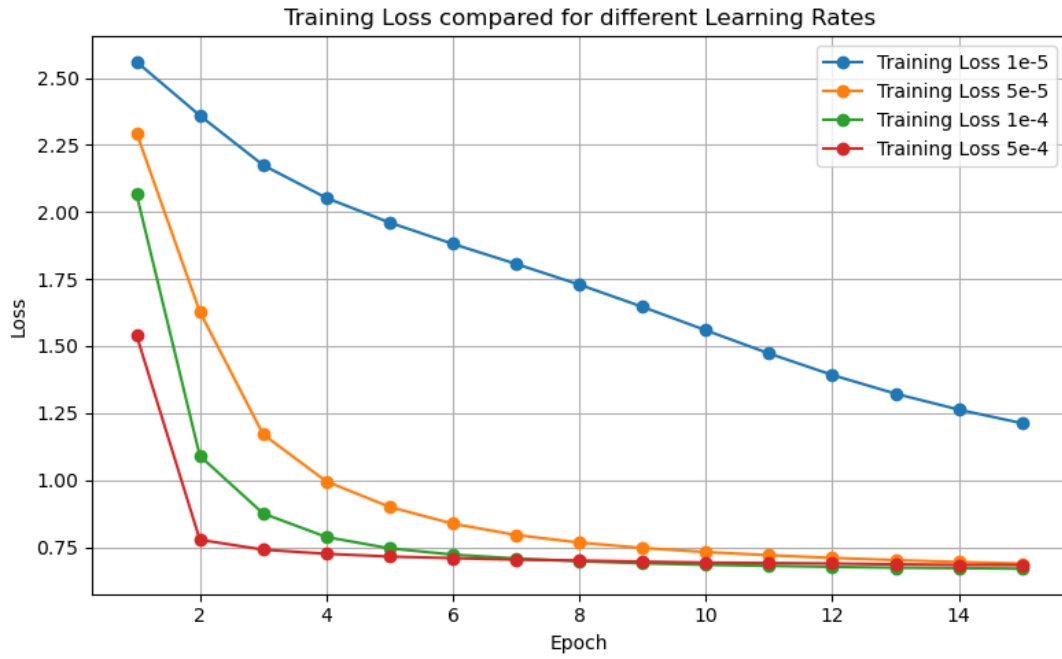


Figure 5.1: Training loss curves for different learning rates. Although the red curve (Learning Rate = 5×10^{-4}) converges faster, the green curve (Learning Rate = 1×10^{-4}) converged to a slightly lower value at the end of training.

was to determine whether increasing model depth leads to better transcription accuracy and lower Cross-Entropy Loss.

The performance results, summarized in Table 5.3, show that increasing the number of layers generally improves the CER and WER and leads to a lower Cross-Entropy Loss, implying that a deeper model captures richer linguistic and phonetic representations.

Model Depth	CER ↓	WER ↓	Cross-Entropy Loss ↓
6 layers	8%	18%	0.09
8 layers	6%	17%	0.06
12 layers	4%	16%	0.05

Table 5.3: ASR performance across different model depths.

Based on these experiments, the final model for Spanish fine-tuning was trained on LibriSpeech 960 hours, with 12 layers and 8 heads, to match HuBERT, and used a learning rate of 1×10^{-4} . Figure 5.2 shows the training and validation loss curves for this model’s training.

5.2 Spanish Fine-Tuning Results

After training the decoder-only transformer model on 960 hours of English data, the model was fine-tuned on varying amounts of Spanish data to evaluate how data availability influences the performance of the model on cross-lingual transfer learning.

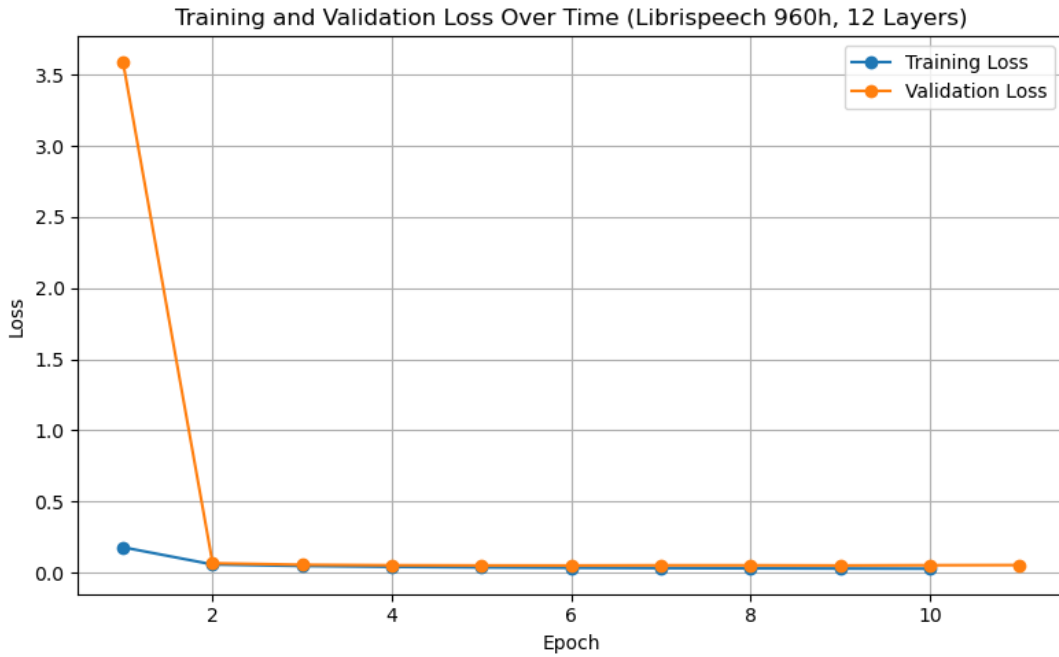


Figure 5.2: Plot of training v/s validation loss curves of the decoder-only transformer model on LibriSpeech 960 hours, with 12 layers and 8 heads. The training loss curve converges to a low value of 0.02 in the last epoch. The validation loss is used to monitor overfitting. The steady convergence of the validation loss curve indicates that the model does not overfit.

5.2.1 Effect of Data on Spanish Fine-Tuning

The model was fine-tuned on 9, 30, 60, 120, and 240 hours of Spanish data. This selection was based on the structure of the Multilingual LibriSpeech dataset, where each partition consists of approximately 7,000 utterances, corresponding to roughly 30 hours of speech. For each fine-tuning stage, the data was doubled till the model stopped overfitting. Similar to English pre-training, each data partition was split into the train-val-test split. The results of the experiments are tabulated in Table 5.4.

Fine-tuning with only 9 hours of Spanish data results in a high CER of 12%, demonstrating that minimal exposure to Spanish yields poor transcription accuracy. This is further supported by the loss curves in Figure 5.3, which indicate signs of overfitting. Initially, both training and validation losses decrease rapidly, suggesting that the model is learning useful features. However, after around the sixth epoch, the validation loss begins to slightly increase, while the training loss continues to decline. This divergence implies that the model is memorizing the training data rather than generalizing well to unseen Spanish speech.

A similar pattern is observed for models fine-tuned on 30 and 60 hours of Spanish data. While these models benefit from larger datasets, they also exhibit overfitting tendencies, particularly in the later epochs. The training loss continues to decrease steadily, while the validation increases, indicating that the model is learning dataset-specific patterns rather than generalizing to new Spanish speech. However, the models fine-tuned on 120

and 240 hours demonstrate better generalization, as reflected in their improved CER and WER values as seen in Table 5.4.

These findings suggest that while small amounts of Spanish data lead to overfitting, fine-tuning on at least 100–250 hours of low-resource language data provides sufficient coverage for the model to generalize effectively.

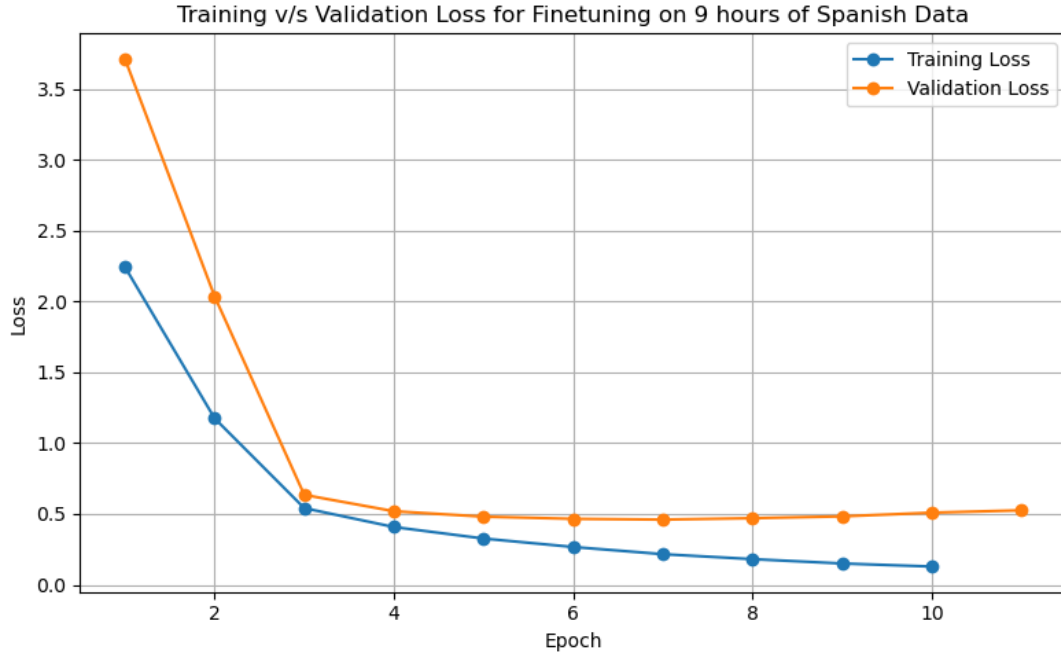


Figure 5.3: Training and validation loss curves compared for fine-tuning the model on 9 hours of Spanish ASR data. The orange validation curve starts to increase after epoch 6, as the blue training loss curve continues to decrease, indicating that the model is overfitting on the training data.

Fine-Tuning Data	CER ↓	WER ↓	Cross-Entropy Loss ↓
30h	10%	46%	0.5
60h	8%	40%	0.4
120h	7%	35%	0.3
240h	6%	28%	0.2

Table 5.4: Effect of Spanish fine-tuning data on ASR performance.

5.2.2 Baseline Comparison

To assess whether decoder-only transformers can effectively perform cross-lingual transfer learning for ASR, the fine-tuned model was compared against the HuBERT model, fine-tuned on Spanish ASR with CTC loss. To ensure an accurate comparison, HuBERT was first fine-tuned on 960 hours of English ASR data from LibriSpeech, followed by 240 hours of Spanish ASR data. The HuBERT model is fine-tuned on

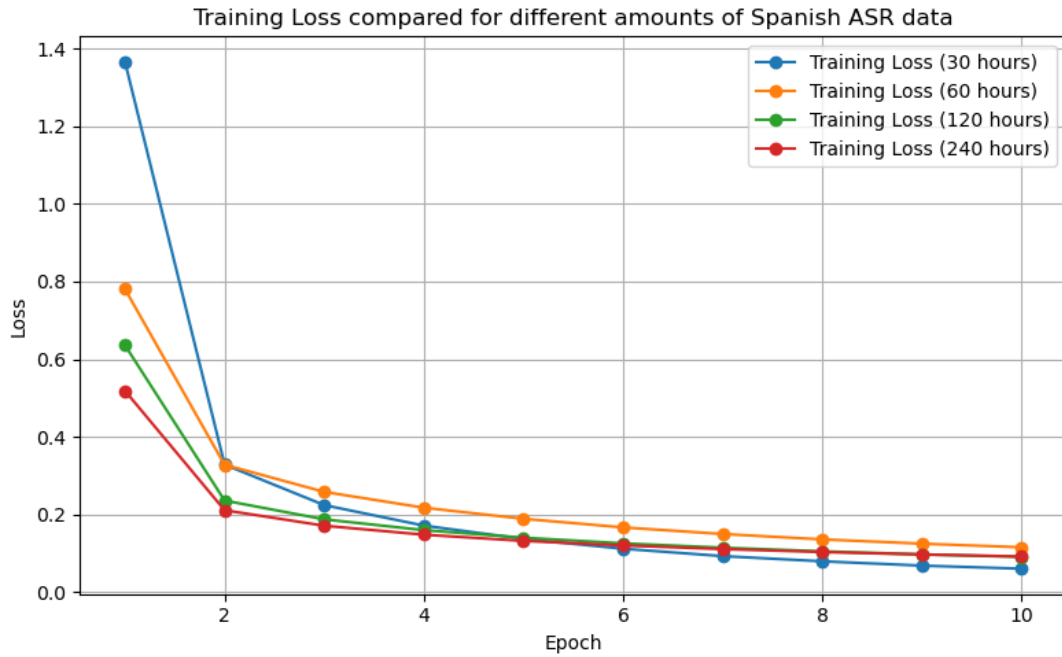


Figure 5.4: Training loss curves for different Spanish fine-tuning dataset sizes.

Model	CER ↓	WER ↓
HuBERT + Spanish (240 hours)	10%	37%
Fine-Tuned English Decoder-Only (240 hours)	6%	28%

Table 5.5: Baseline comparison for Spanish ASR.

Spanish ASR data for the same number of epochs as the decoder-only transformer model. The results are tabulated in Table 5.5.

The results demonstrate that the decoder-only transformer, when fine-tuned on 240 hours of Spanish data, achieves a lower Character Error Rate (6%) and Word Error Rate (28%) compared to the HuBERT baseline (10% CER, 37% WER). This suggests that decoder-only architectures, when pre-trained on high-resource languages like English, can effectively transfer learned representations to low-resource languages.

5.3 Discussion of Findings

The results of the experiments demonstrate that pre-trained decoder-only models can effectively transfer knowledge across languages, achieving not just competitive but superior performance in cross-lingual transfer learning for ASR. As shown in Table 5.5, the decoder-only model achieves a 40% relative reduction in CER and 24% in WER compared to the HuBERT baseline, suggesting advantages in using this architecture. The experiments indicate that cross-lingual adaptation stabilizes significantly when the fine-tuning dataset reaches approximately 100 hours, which is a lot smaller than the pre-trained 960 hours of English data. These findings highlight the potential of decoder-only architectures as a robust solution for ASR in low-resource languages.

Chapter 6

Conclusions

This chapter presents the concluding discussion of the dissertation, reflecting on the key findings and their broader implications. It begins by summarizing the main contributions of the project in the context of the decoder-only transformer architecture and cross-lingual transfer learning for ASR. This is followed by an outline of the limitations encountered during the research, which highlights potential areas for improvement. Finally, the chapter concludes with a discussion of directions for future work.

6.1 Summary of Contributions

This dissertation investigated the potential of decoder-only transformer architectures for cross-lingual transfer learning in ASR. Through a series of experiments, the impact of training data size, model depth, and learning rate on the performance of the English ASR model was assessed. These experiments established that increasing the training data and the depth of the model substantially improved the CER and WER of the transcriptions.

Following the development of the pre-trained English decoder, the model was fine-tuned on varying amounts of Spanish ASR data to evaluate its cross-lingual transfer abilities. This showed promising results, as the Spanish ASR model achieved competitive CER and WER despite being trained on limited Spanish data. This highlights the viability of decoder-only transformer models for efficient cross-lingual adaptation for low-resource languages.

As the interest in integrating Large Language Models with speech grows, this project highlights that decoder-only transformer architectures can serve as an effective bridge between language understanding and speech recognition. Their ability to adapt across similar languages with limited supervision makes them well-suited for multi-modal, multilingual applications. This positions decoder-only models as promising candidates for future end-to-end systems that unify speech and text processing under a single architecture.

6.2 Limitations

Despite its promising results, this project has some limitations. First, the evaluation was limited to a single language pair using English as the source language and Spanish as the target language. While Spanish shares some phonetic and orthographic similarities with English, it may not fully represent the challenges of cross-lingual transfer to languages with greater linguistic divergence, such as those with non-Latin scripts, tonal systems, or significantly different phonological (sound) structures. This limits the generalizability of the findings across the approximately 7000 languages worldwide.

Second, although this study employs a decoder-only transformer architecture, it relies on a pre-trained HuBERT encoder as a frozen feature extractor to process raw speech waveforms into contextualized embeddings. In contrast to traditional encoder-decoder transformers, where the encoder is trained and dynamically learns acoustic representations alongside the decoder, the HuBERT encoder, in this implementation, remains static, serving only to extract features. This design choice aligns with the goal of simplifying the architecture by concentrating all learnable parameters within the decoder, which could enhance computational efficiency by reducing the overhead of training a full encoder-decoder system. However, the actual efficiency of this decoder-only approach, particularly in terms of training time, inference speed, and resource utilization, remains undetermined. The dependence on a pre-trained, frozen encoder like HuBERT, with its 95 million parameters, may affect the anticipated efficiency, as the feature extraction step still requires significant computational cost. Without a detailed comparative analysis of runtime and resource demands against fully trainable encoder-decoder models, the efficiency of the decoder-only transformer architecture requires further investigation.

6.3 Future Work

The results of this project open several avenues for future research. One key direction would be to evaluate the computational efficiency of decoder-only transformer models. While this architecture reduces the number of trainable components by offloading feature extraction to a frozen HuBERT encoder, the overall system may still suffer from significant computational costs. Future research can assess training and inference time, memory usage, and scalability compared to fully encoder-decoder models, helping determine the efficiency of decoder-only architectures.

Another important direction is to extend the cross-lingual transfer experiments to more typologically diverse language pairs. Although Spanish is linguistically different from English, it shares a similar script and certain phonological features. Exploring target languages with different scripts (e.g., Devanagari, Arabic), tonal systems (e.g., Mandarin), or morphosyntactic structures (e.g., Finnish or Turkish) would provide a better understanding of the generalizability of decoder-only models for cross-lingual transfer learning in ASR. This could also be extended to a multilingual setting, where a single decoder is trained or fine-tuned across multiple target languages simultaneously.

Bibliography

- Divyanshu Aggarwal, Ashutosh Sathe, and Sunayana Sitaram. Exploring pretraining via active forgetting for improving cross lingual transfer for decoder language models. *arXiv preprint arXiv:2410.16168*, 2024.
- Wasi Uddin Ahmad, Zhisong Zhang, Xuezhe Ma, Eduard Hovy, Kai-Wei Chang, and Nanyun Peng. On difficulties of cross-lingual transfer with order differences: A case study on dependency parsing. *arXiv preprint arXiv:1811.00570*, 2018.
- Rosana Ardila, Megan Branson, Kelly Davis, Michael Henretty, Michael Kohler, Josh Meyer, Reuben Morais, Lindsay Saunders, Francis M Tyers, and Gregor Weber. Common voice: A massively-multilingual speech corpus. *arXiv preprint arXiv:1912.06670*, 2019.
- Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. way2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in neural information processing systems*, 33:12449–12460, 2020.
- R Davis, NS Sutherland, and BR Judd. Information content in recognition and recall. *Journal of Experimental Psychology*, 61(5):422, 1961.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pages 4171–4186, 2019.
- Linus Ericsson, Henry Gouk, Chen Change Loy, and Timothy M Hospedales. Self-supervised representation learning: Introduction, advances, and challenges. *IEEE Signal Processing Magazine*, 39(3):42–62, 2022.
- Carlos Escolano, Francesca De Luca Fornaciari, and Maite Melero. Residual dropout: A simple approach to improve transformer’s data efficiency. In *The 3rd Annual Meeting of the ELRA-ISCA Special Interest Group on Under-resourced Languages@ LREC-COLING-2024 (SIGUL 2024): Turin, Italy*, pages 294–299. ELRA Language Resources Association and the International Committee on . . . , 2024.
- Ankit Gupta, George Saon, and Brian Kingsbury. Exploring the limits of decoder-

- only models trained on public speech recognition corpora. *arXiv preprint arXiv:2402.00235*, 2024.
- Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM transactions on audio, speech, and language processing*, 29:3451–3460, 2021.
- Qingyang Li and Weimao Ke. Investigating the synergistic effects of dropout and residual connections on language model training. *arXiv preprint arXiv:2410.01019*, 2024.
- Benjamin Minixhofer, Fabian Paischer, and Navid Rekabsaz. Wechsel: Effective initialization of subword embeddings for cross-lingual transfer of monolingual language models. *arXiv preprint arXiv:2112.06598*, 2021.
- Vinod Nair and Geoffrey E Hinton. Rectified linear units improve restricted boltzmann machines. In *Proceedings of the 27th international conference on machine learning (ICML-10)*, pages 807–814, 2010.
- Xueyan Niu, Bo Bai, Lei Deng, and Wei Han. Beyond scaling laws: Understanding transformer performance with associative memory. *arXiv preprint arXiv:2405.08707*, 2024.
- Myle Ott, Sergey Edunov, Alexei Baevski, Angela Fan, Sam Gross, Nathan Ng, David Grangier, and Michael Auli. fairseq: A fast, extensible toolkit for sequence modeling. In *Proceedings of NAACL-HLT 2019: Demonstrations*, 2019.
- Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: an asr corpus based on public domain audio books. In *2015 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 5206–5210. IEEE, 2015.
- Douglas B. Paul and Janet M. Baker. The design for the Wall Street Journal-based CSR corpus. In *Speech and Natural Language: Proceedings of a Workshop Held at Harriman, New York, February 23-26, 1992*, 1992. URL <https://aclanthology.org/H92-1073/>.
- Leena G Pillai, Kavya Manohar, Basil K Raju, and Elizabeth Sherly. Multistage fine-tuning strategies for automatic speech recognition in low-resource languages. *arXiv preprint arXiv:2411.04573*, 2024.
- Emiel Por, Maaïke van Kooten, and Vanja Sarkovic. Nyquist–shannon sampling theorem. *Leiden University*, 1(1):1–2, 2019.
- Vineel Pratap, Qiantong Xu, Anuroop Sriram, Gabriel Synnaeve, and Ronan Collobert. Mls: A large-scale multilingual dataset for speech research. *arXiv preprint arXiv:2012.03411*, 2020.
- Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. Improving language understanding by generative pre-training. 2018.

- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023.
- Andrew Rouditchenko, Sameer Khurana, Samuel Thomas, Rogerio Feris, Leonid Karlinsky, Hilde Kuehne, David Harwath, Brian Kingsbury, and James Glass. Comparison of multilingual self-supervised and weakly-supervised speech pre-training for adaptation to unseen languages. *arXiv preprint arXiv:2305.12606*, 2023.
- Anthony Rousseau, Paul Deléglise, and Yannick Esteve. Ted-lium: an automatic speech recognition dedicated corpus. In *LREC*, pages 125–129, 2012.
- Josh Starmer. Coding a chatgpt like transformer from scratch in pytorch, Jul. 2024. URL <https://www.youtube.com/watch?v=C9QSp15nmrY>.
- Emiru Tsunoo, Hayato Futami, Yosuke Kashiwagi, Siddhant Arora, and Shinji Watanabe. Decoder-only architecture for speech recognition with ctc prompts and text data augmentation. *arXiv preprint arXiv:2309.08876*, 2023.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- Valentin Vielzeuf. Investigating the ‘autoencoder behavior’ in speech self-supervised models: a focus on hubert’s pretraining. *arXiv preprint arXiv:2405.08402*, 2024.
- Changhan Wang, Juan Pino, and Jiatao Gu. Improving cross-lingual transfer learning for end-to-end speech recognition with speech translation. *arXiv preprint arXiv:2006.05474*, 2020.
- Thomas Wang, Adam Roberts, Daniel Hesslow, Teven Le Scao, Hyung Won Chung, Iz Beltagy, Julien Launay, and Colin Raffel. What language model architecture and pretraining objective works best for zero-shot generalization? In *International Conference on Machine Learning*, pages 22964–22984. PMLR, 2022.
- Ronald J Williams and David Zipser. A learning algorithm for continually running fully recurrent neural networks. *Neural computation*, 1(2):270–280, 1989.
- Dong Zhang, Shimin Li, Xin Zhang, Jun Zhan, Pengyu Wang, Yaqian Zhou, and Xipeng Qiu. Speechgpt: Empowering large language models with intrinsic cross-modal conversational abilities. *arXiv preprint arXiv:2305.11000*, 2023.
- Jing Zhao and Wei-Qiang Zhang. Improving automatic speech recognition performance for low-resource languages with self-supervised models. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1227–1241, 2022.